
Security Review Report
NM-0695-B Origami Finance



NETHERMIND
SECURITY

(December 01, 2025)

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1 Executive Summary

This document presents the results of a security review conducted by **Nethermind Security** for **Origami Finance** OPAL contracts.

OPAL (Origami Portfolio of Assets and Liabilities) is a tokenized balance sheet vault that issues a single ERC-20 share representing a managed portfolio of on-chain assets and liabilities. It tracks both assets and liabilities within a single token.

The core components in scope for this audit are the `OpalVault` and its `OpalManager`, alongside the set of adapters. The vault inherits from `OrigamiTokenizedBalanceSheetVault` and serves as the main entry point for users, exposing functions to join or exit the vault. The manager acts as the aggregation layer, orchestrating multiple adapters and allocating assets across them.

Adapters are modular contracts that interface with specific underlying protocols or hold spot assets. Current adapters include integrations for Aave V3, Euler V2 vaults, Morpho markets, and a spot asset adapter for ERC-20 tokens. They are deployed as minimal, gas-efficient clones with immutable arguments and are controlled through a whitelisted factory.

The audit comprises 2454 lines of the Solidity code. **The audit was performed using** (a) manual analysis of the codebase, and (b) automated analysis tools.

Along this document, we report 4 points of attention, classified as Informational or Best Practices severity. The issues are summarized in Fig. 1.

This document is organized as follows. Section 2 presents the files in the scope. Section 3 summarizes the issues. Section 4 presents the system overview. Section 5 discusses the risk rating methodology. Section 6 details the issues. Section 7 discusses the documentation provided by the client for this audit. Section 8 presents the test suite evaluation and automated tools used. Section 9 concludes the document.

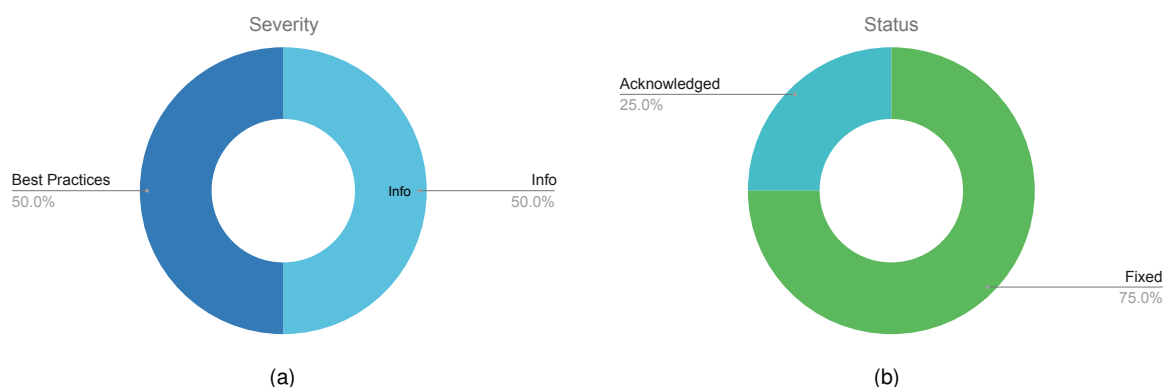


Fig. 1: Distribution of issues: Critical (0), High (0), Medium (0), Low (0), Undetermined (0), Informational (2), Best Practices (2).
Distribution of status: Fixed (3), Acknowledged (1), Mitigated (0), Unresolved (0)

Summary of the Audit

Audit Type	Security Review
Initial Report	November 28, 2025
Final Report	December 01, 2025
Initial Commit	dca61cba9a8dd67ada44c90ec2f96264bce86860
Final Commit	48319d0b13e3037bd5f98c05825f763d12eb0916
Documentation Assessment	High
Test Suite Assessment	High

2 Audited Files

	Contract	LoC	Comments	Ratio	Blank	Total
1	OpalAdapterFactory.sol	74	17	23.0%	17	108
2	OpalVault.sol	250	103	41.2%	62	415
3	OpalManager.sol	590	229	38.8%	122	941
4	adapters/OpalAdapterAaveV3.sol	498	125	25.1%	122	745
5	adapters/OpalAdapterSpotAssets.sol	138	33	23.9%	34	205
6	adapters/OpalAdapterEuler.sol	398	165	41.5%	116	679
7	adapters/OpalAdapterMorpho.sol	404	118	29.2%	102	624
8	adapters/OpalAdapterBase.sol	102	50	49.0%	33	185
	Total	2454	840	34.2%	608	3902

3 Summary of Issues

	Finding	Severity	Status
1	Missing supportsInterface overrides in Opal Adapter contracts	Info	Fixed
2	Potential loss of accrued performance fee	Info	Acknowledged
3	Lack of checks for addresses in adapters against the markets.	Best Practices	Fixed
4	Improper handling of "Unlimited caps" during fee adjustments	Best Practices	Fixed

4 System Overview

The OPAL (Origami Portfolio Assets & Liabilities) protocol provides users with a structured mechanism for participation in a professionally managed portfolio. The protocol's pooled funds are constituted of assets and corresponding liabilities strategically deployed across multiple diverse financial markets. User participation is facilitated through the holding of a share of the portfolio, granting proportional exposure to its net value. The protocol is engineered with a highly modular architecture, which allows for the seamless integration and utilization of new external protocols through the introduction of specialized adapter contracts.

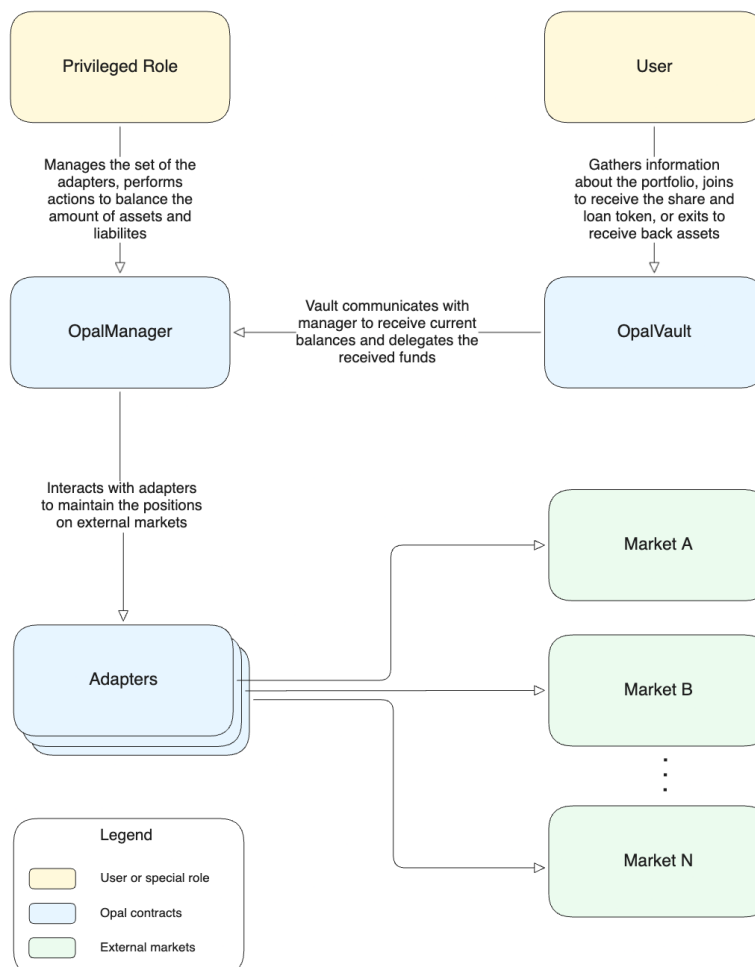


Fig. 2: Main Opal components

4.1 Opal Vault and Opal Manager

The `OpalVault` is the implementation of a Balance Sheet Vault, where shares represent a proportional slice of the vault's consolidated balance sheet. This balance sheet comprises assets, which represent the vault's positive value, and liabilities, which represent the debt obligation owed by the vault. Consequently, a user holding a share participates in a pool of assets and associated debt managed by the vault. The `OpalVault` is linked to the `OpalManager`, enabling the effective utilization of assets and the active management of debt across multiple external markets.

The `OpalVault` serves as the main external entry point for user interaction. Users can acquire the vault's shares by invoking the `joinWithToken()` function, where the amount of a specific token is specified, or by calling `joinWithShares()`, allowing users to define the particular amount of shares desired during the joining action. The share inherently represents the managed assets and liabilities. Upon user entry, the user must provide proportional amounts of required collateral assets, which are subsequently passed to the `OpalManager`. In return, the user receives the vault's shares and the corresponding liabilities tokens. Conversely, upon an exit action, the reverse process is performed: the liabilities are transferred from the user, and the proportional collateral is returned. The `OpalVault` exposes crucial query functions that provide informative data to the user, such as the maximum current join capacity or the computed amounts obtained during simulated join and exit operations.

The valuation, or price, of the `OpalVault` share is defined during the initial seeding. This initial seeding is performed by a privileged role and is necessary for defining the initial proportional amounts of each asset and liability. These proportions may subsequently evolve throughout the vault's operational lifecycle through strategic balancing activities.

Core functions exposed by the `OpalVault`:

- **seed(...)**: Called by the privileged role to determine the initial share price of each asset and liability.
- **joinWithToken(...)** / **joinWithShares(...)**: Called by the user to buy a vault's shares given a specified amount of a particular token or shares.
- **exitWithToken(...)** / **exitWithShares(...)**: Called by the user to burn shares and receive the supplied assets.
- **previewJoinWithToken(...)** / **previewJoinWithShares(...)**: Simulates the effects of the join at the current block.
- **previewExitWithToken(...)** / **previewExitWithShares(...)**: Simulates the effects of the exit at the current block.
- **maxJoinWithToken(...)** / **maxJoinWithShares(...)**: The maximum amount of tokens that can be deposited to the vault, or the amount of shares that can be minted.
- **maxExitWithToken(...)** / **maxExitWithShares(...)**: The maximum amount of tokens that can be withdrawn from the vault or the vault's shares to be redeemed.

The `OpalManager` is primarily responsible for the governance and management of the adapters. An adapter is an essential contract required for secure interaction with external markets. The `OpalManager` allows the privileged role to securely introduce a new adapter or to remove an existing one. Its responsibility is the allocation of assets and liabilities across all registered adapters. Furthermore, the `OpalManager` provides critical information regarding the aggregated assets and liabilities across all adapters to the `OpalVault`, ensuring the accurate and correct calculation of the share price.

The `OpalManager` exposes a specialized multicall function, which facilitates the execution of privileged actions on the integrated adapters. This crucial functionality empowers the designated owner (special role) to conduct sophisticated rebalancing of the vault's capital structure, including operations such as borrowing, repaying debt, or supplying assets. These management actions are performed independently of the user-facing join and exit flows, ensuring strategic control and dynamic portfolio optimization.

Core functions exposed by the `OpalManager`:

- **addAdapter(...)**: Creates a new adapter for a given implementation via the adapter factory and adds it to the aggregated balance sheet.
- **removeAdapter(...)**: Removes an adapter from the aggregated set.
- **balanceSheet(...)**: Aggregates the balance sheet data across all the adapters.
- **join(...)**: Called by the vault to add equity by allocating the assets and borrowing the liabilities.
- **exit(...)**: Called by the vault to remove equity by repaying 'liabilities' and removing 'assets' collateral from the underlying adapters.
- **multicall(...)**: Allows privileged caller to perform calls from the `OpalManager`.

4.2 OPAL Adapters

Adapters are modular contracts that interface with specific underlying protocols or hold spot assets. They are deployed through the `OpalAdapterFactory` as minimal, gas-efficient clones with immutable arguments and are controlled via a whitelisted factory. Adapter creation is performed by the `OpalManager` using the `create(...)` function.

4.2.1 Core Functions

Each adapter exposes core functionalities for asset and liability management:

- **join(...)**: Called by the manager to allocate assets into the underlying protocol (if any) and deliver the corresponding liabilities to the recipient.
- **exit(...)**: Called by the manager to withdraw assets from the underlying protocol and repay liabilities to the recipient.
- **merklClaim(...)**: Claims rewards from the Merkl rewards distributor for adapters that use the Merkl rewards system.
- **balanceSheet()**: This function returns the adapter's current asset and liability holdings, it is crucial for the manager to calculate accurate allocations across adapters.
- **maxJoin** / **maxExit**: Return the maximum amounts of assets and liabilities that can be joined or exited, reflecting the underlying protocol's limits. Returns 0 if the market is disabled or frozen.

Adapters also provide health and risk metrics through the following functions:

- **currentLtv()**: Calculates the current loan-to-value ratio of the adapter position.
- **liquidationLtv()**: Returns the liquidation threshold from the underlying protocol.
- **healthFactor()**: Calculated based on current and liquidation LTV.

Additionally, a Loan utilization ratio is computed per adapter to reflect the underlying protocol state, it is computed as follows:

$$\text{LoanTokenUtilizationRatio} = \frac{\text{alreadyBorrowed}}{\text{totalDebtSupply}} \quad (1)$$

Where `alreadyBorrowed` is the total borrowed amount in the protocol, and `totalDebtSupply` includes both borrowed and available debt capacity. This ratio is checked against a protocol specified max value after each borrow operation.

4.2.2 Adapter Types

Different adapters have been implemented, interacting with different underlying protocols.

- **Morpho Adapter:** This adapter manages a borrow position in a single Morpho market, with one asset token and one liability token. In the join flow, assets are supplied as collateral and liabilities are borrowed on behalf of the user. In the exit flow, liabilities are repaid, and the collateral is withdrawn. The adapter enforces a safe LTV that is set below Morpho's liquidation threshold as a precautionary measure.
- **Aave V3 Adapter:** The Aave V3 adapter manages a borrow position in an Aave V3 market, with one debt token and one or more collateral tokens. During the join flow, asset tokens are supplied to the Aave pool, and liabilities are subsequently borrowed and sent to the user. The exit flow reverses this process, repaying liabilities and withdrawing collateral.
- **Euler V2 Adapter:** The Euler adapter manages a borrow position in an Euler V2 market with a single asset token and a single liability token. It interacts with Euler's `collateralVault` and `debtVault`, supplying and withdrawing collateral from the former and borrowing or repaying debt via the latter.
- **Spot Assets Adapter:** The Spot Assets adapter holds fixed ERC-20 asset tokens and does not manage liabilities. Joining the adapter consists of transferring ERC-20 tokens into it, while exiting allows the withdrawal of those tokens.

In addition to the main flows, all adapters expose bundler-only functions that allow direct interaction with underlying protocols, such as supplying collateral, borrowing, repaying, or withdrawing. These functions enable the bundler to perform rebalancing or adapter migration as needed.

5 Risk Rating Methodology

The risk rating methodology used by [Nethermind Security](#) follows the principles established by the [OWASP Foundation](#). The severity of each finding is determined by two factors: **Likelihood** and **Impact**.

Likelihood measures how likely the finding is to be uncovered and exploited by an attacker. This factor will be one of the following values:

- a) **High**: The issue is trivial to exploit and has no specific conditions that need to be met;
- b) **Medium**: The issue is moderately complex and may have some conditions that need to be met;
- c) **Low**: The issue is very complex and requires very specific conditions to be met.

When defining the likelihood of a finding, other factors are also considered. These can include but are not limited to motive, opportunity, exploit accessibility, ease of discovery, and ease of exploit.

Impact is a measure of the damage that may be caused if an attacker exploits the finding. This factor will be one of the following values:

- a) **High**: The issue can cause significant damage, such as loss of funds or the protocol entering an unrecoverable state;
- b) **Medium**: The issue can cause moderate damage, such as impacts that only affect a small group of users or only a particular part of the protocol;
- c) **Low**: The issue can cause little to no damage, such as bugs that are easily recoverable or cause unexpected interactions that cause minor inconveniences.

When defining the impact of a finding, other factors are also considered. These can include but are not limited to Data/state integrity, loss of availability, financial loss, and reputation damage. After defining the likelihood and impact of an issue, the severity can be determined according to the table below.

		Severity Risk		
Impact	High	Medium	High	Critical
	Medium	Low	Medium	High
	Low	Info/Best Practices	Low	Medium
	Undetermined	Undetermined	Undetermined	Undetermined
		Low	Medium	High
		Likelihood		

To address issues that do not fit a High/Medium/Low severity, [Nethermind Security](#) also uses three more finding severities: **Informational**, **Best Practices**, and **Undetermined**.

- a) **Informational** findings do not pose any risk to the application, but they carry some information that the audit team intends to pass to the client formally;
- b) **Best Practice** findings are used when some piece of code does not conform with smart contract development best practices;
- c) **Undetermined** findings are used when we cannot predict the impact or likelihood of the issue.

6 Issues

6.1 [Info] Missing supportsInterface overrides in Opal Adapter contracts

File(s): `OpalAdapterBase.sol`, `OpalAdapterAaveV3.sol`, `OpalAdapterEuler.sol`, `OpalAdapterMorpho.sol`, `OpalAdapterSpotAssets.sol`

Description: The `OrigamiBundlerPluginCore` contract inherits from `IERC165` and implements `supportsInterface`, declaring support for both `IOrigamiBundlerPlugin` and `IERC165` interfaces.

However, the different Opal adapter contracts extend `OrigamiBundlerPluginCore` but do not override `supportsInterface` to include the additional interfaces they support. As a result, external callers querying the interface support will not recognize the adapter contracts as implementing `IOpalAdapter` and the specific adapter interfaces.

Recommendation(s): Consider overriding `supportsInterface` in each adapter implementation to include support for `IOpalAdapter` and different adapter-specific interfaces.

Status: Fixed

Update from Nethermind Security: This issue was self-identified by the Origami team during the audit.

Update from the client: Fixed in commit [377b379c366e3adbb1ad8e873004eaa62f380281](#)

6.2 [Info] Potential loss of accrued performance fee

File(s): `OpalVault.sol`

Description: The `_collectPerformanceFees()` function does not mint fee shares if the calculated fee amount is zero. However, it still updates the `lastPerformanceFeeTime`, effectively resetting the fee accrual window. This causes small accumulated fee amounts to be lost.

```
1 function _collectPerformanceFees() internal returns (uint256 amount) {
2     amount = accruedPerformanceFee();
3     if (amount != 0) {
4         address _feeCollector = feeCollector;
5         emit PerformanceFeesCollected(_feeCollector, amount);
6
7         // Do not need to check vs maxTotalSupply here as it is
8         // only for new user investments
9         _mint(_feeCollector, amount);
10    }
11    // @audit: Update without the mint
12    lastPerformanceFeeTime = uint48(block.timestamp);
13 }
```

Recommendation(s): Consider reverting when `amount == 0` to avoid losing dust fees.

Status: Acknowledged

Update from the client: This is an elevated access only function, so I think its sensible for the elevated access to choose if its sensible to call the function (and reset the performance fee time) or wait. I dont think this warrants a finding or changing the behaviour in the contract to prevent it from being possible.

6.3 [Best Practice] Lack of checks for addresses in adapters against the markets.

File(s): [OpalAdapterMorpho.sol](#), [OpalAdapterAaveV3.sol](#)

Description: The OpalAdapterMorpho contract initializes its configuration through `_adapterInit()`, which sets `_collateralToken` and `_loanToken`. However, these addresses are not validated against the actual Morpho market configuration available via the `idToMarketParams` mapping. The lack of this validation may cause incorrect initialization of the adapter. Note that those checks are present during initialization in the other adapters.

Additionally, the `OpalAdapterAaveV3.updateAavePool()` function does not verify that the adapter's loan and collateral tokens match the reserve configuration returned by `_aavePool.getReserveData()`.

Recommendation(s): Consider adding validation to ensure that `_collateralToken` and `_loanToken` match the market's configuration in both Morpho and Aave V3.

Status: Fixed

Update from the client: OK yes they are a sensible additions to both `OpalAdapterMorpho` and `OpalAdapterAaveV3`, will add.

Fixed in commit [629f4bdb750ac65e5bd14f60b1907e3231b15905](#)

6.4 [Best Practices] Improper handling of "Unlimited caps" during fee adjustments

File(s): [OpalVault.sol](#)

Description: The internal functions `_leastOfMaxExitWithShares()` and `_leastOfMaxJoinWithShares()` in `OpalManager` determine the maximum number of shares permitted across all adapters and tokens. When no explicit join or exit caps are defined, the logic uses `type(uint256).max` as a value to represent unlimited capacity.

However, the current implementation applies fee adjustments to this value without first checking whether it represents the "no cap" case. Specifically:

- In `_leastOfMaxExitWithShares()`: The code calls `fees.inverseSubtractFee()`, which will overflow and revert when trying to add fees to `type(uint256).max`;
- In `_leastOfMaxJoinWithShares()`: The code calls `fees.subtractFee()` which could return incorrect values when operating on `type(uint256).max`;

```

1  function _leastOfMaxExitWithShares(...) private view returns (uint256 maxShares) {
2      {
3          (uint256[] memory maxAssets, uint256[] memory maxLiabilities) = _manager.maxExit();
4
5          maxShares = _maxSharesFromTokens($.assetBalances, maxAssets, $.totalSupply, type(uint256).max);
6          maxShares = _maxSharesFromTokens($.liabilityBalances, maxLiabilities, $.totalSupply, maxShares);
7
8          // @audit maxShares could be `type(uint256).max` if no caps are implemented
9          maxShares = maxShares.inverseSubtractBps(exitFeeBps(), OrigamiMath.Rounding.ROUND_UP);
10     }
11
12     uint256 maxSharesGivenTotalSupplyCap = _maxExitWithShares(sharesOwner);
13     if (maxSharesGivenTotalSupplyCap < maxShares) maxShares = maxSharesGivenTotalSupplyCap;
14 }

```

This issue currently affects view functions only and is unlikely to manifest in practice, since `maxShares` is subsequently constrained by the user's actual share balance in `_maxExitWithShares()` and by total supply limitations. Nevertheless, the current behavior is misleading and may produce unexpected results.

Recommendation(s): Introduce an explicit guard to detect the `type(uint256).max` value before applying fee adjustments. If the value indicates "no cap," skip fee calculations.

Status: Fixed

Update from the client: Ack - we will fix that and add more tests to cover. It should have a `if (maxShares < type(uint256).max)` prefix like in `maxJoinWithToken()`.

Fixed in commit: [48319d0b13e3037bd5f98c05825f763d12eb0916](#)

7 Documentation Evaluation

Software documentation refers to the written or visual information that describes the functionality, architecture, design, and implementation of software. It provides a comprehensive overview of the software system and helps users, developers, and stakeholders understand how the software works, how to use it, and how to maintain it. Software documentation can take different forms, such as user manuals, system manuals, technical specifications, requirements documents, design documents, and code comments. Software documentation is critical in software development, enabling effective communication between developers, testers, users, and other stakeholders. It helps to ensure that everyone involved in the development process has a shared understanding of the software system and its functionality. Moreover, software documentation can improve software maintenance by providing a clear and complete understanding of the software system, making it easier for developers to maintain, modify, and update the software over time. Smart contracts can use various types of software documentation. Some of the most common types include:

- Technical whitepaper: A technical whitepaper is a comprehensive document describing the smart contract's design and technical details. It includes information about the purpose of the contract, its architecture, its components, and how they interact with each other;
- User manual: A user manual is a document that provides information about how to use the smart contract. It includes step-by-step instructions on how to perform various tasks and explains the different features and functionalities of the contract;
- Code documentation: Code documentation is a document that provides details about the code of the smart contract. It includes information about the functions, variables, and classes used in the code, as well as explanations of how they work;
- API documentation: API documentation is a document that provides information about the API (Application Programming Interface) of the smart contract. It includes details about the methods, parameters, and responses that can be used to interact with the contract;
- Testing documentation: Testing documentation is a document that provides information about how the smart contract was tested. It includes details about the test cases that were used, the results of the tests, and any issues that were identified during testing;
- Audit documentation: Audit documentation includes reports, notes, and other materials related to the security audit of the smart contract. This type of documentation is critical in ensuring that the smart contract is secure and free from vulnerabilities.

These types of documentation are essential for smart contract development and maintenance. They help ensure that the contract is properly designed, implemented, and tested, and they provide a reference for developers who need to modify or maintain the contract in the future.

Remarks about Origami Finance documentation

The Opal protocol documentation was provided through the README file within the `opal` folder, offering a general overview of the vault and its main functionalities. Additional, detailed documentation was available in the `opal/docs` folder, outlining each component of the system along with its purpose and key functions.

The codebase contains detailed NatSpec comments and inline explanations, describing the intended behavior of the system and the reasoning behind various design choices.

Moreover, the Origami team was proactive during the audit, providing detailed answers to all questions raised by the Nethermind Security team, both during regular calls and through asynchronous communications.

8 Test Suite Evaluation

8.1 Tests Output

```
> forge test --match-path test/foundry/unit/investments/opal/**/*.t.sol

Ran 9 tests for test/foundry/unit/investments/opal/OpalAdapterFactory.t.sol:OpalAdapterFactoryTestAdmin
[PASS] test_addImplementation_failure_alreadyRegistered() (gas: 17817)
[PASS] test_addImplementation_failure_badImpl() (gas: 40036)
[PASS] test_addImplementation_success() (gas: 52301)
[PASS] test_create_failure_notRegistered() (gas: 19873)
[PASS] test_create_success_noArgs() (gas: 82389)
[PASS] test_create_success_withArgs() (gas: 114719)
[PASS] test_initialization() (gas: 27895)
[PASS] test_removeImplementation_failure_notRegistered() (gas: 22127)
[PASS] test_removeImplementation_success() (gas: 25390)
Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 4.69ms (2.26ms CPU time)

Ran 4 tests for test/foundry/unit/investments/opal/OpalAdapterFactory.t.sol:OpalAdapterFactoryTestList
[PASS] test_implementationDetailList_empty() (gas: 14485)
[PASS] test_implementationDetailList_exactlyOnePage() (gas: 65089)
[PASS] test_implementationDetailList_multiPages() (gas: 81692)
[PASS] test_implementationDetailList_underOnePage() (gas: 65152)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 4.87ms (1.34ms CPU time)

Ran 2 tests for test/foundry/unit/investments/opal/OpalAdapterFactory.t.sol:OpalAdapterFactoryTestAccess
[PASS] test_access_addImplementation() (gas: 17678)
[PASS] test_access_removeImplementation() (gas: 17691)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 4.42ms (404.85µs CPU time)

Ran 2 tests for test/foundry/unit/investments/opal/OpalManager.t.sol:OpalManagerTestView
[PASS] test_matchToken() (gas: 3372836)
[PASS] test_supportsInterface() (gas: 9204)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 16.93ms (3.62ms CPU time)

Ran 1 test for test/foundry/unit/investments/opal/OpalManager.t.sol:OpalManagerTestNonImmutableAdapter
[PASS] test_changingAdapter() (gas: 5603814)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 20.79ms (4.77ms CPU time)

Ran 8 tests for test/foundry/unit/investments/opal/OpalManager.t.sol:OpalManagerTestRemoveAdapters
[PASS] test_removeAdapter_fail_empty() (gas: 765670)
[PASS] test_removeAdapter_fail_notDeprecated() (gas: 805143)
[PASS] test_removeAdapter_fail_notFound() (gas: 1303095)
[PASS] test_removeAdapter_success_fromEnd() (gas: 2641110)
[PASS] test_removeAdapter_success_fromMiddle() (gas: 2665345)
[PASS] test_removeAdapter_success_fromStart() (gas: 2660757)
[PASS] test_removeAdapter_success_noAssets() (gas: 802693)
[PASS] test_removeAdapter_success_noLiabilities() (gas: 817412)
Suite result: ok. 8 passed; 0 failed; 0 skipped; finished in 34.10ms (23.21ms CPU time)

Ran 7 tests for test/foundry/unit/investments/opal/OpalManager.t.sol:OpalManagerTestAdmin
[PASS] test_initialization() (gas: 74750)
[PASS] test_recoverToken_success() (gas: 101076)
[PASS] test_setFees_fail() (gas: 45979)
[PASS] test_setFees_success() (gas: 40188)
[PASS] test_setPaused() (gas: 64978)
[PASS] test_setPauser() (gas: 33732)
[PASS] test_setPluginApproved() (gas: 288924)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 36.39ms (3.33ms CPU time)

Ran 12 tests for test/foundry/unit/investments/opal/OpalManager.t.sol:OpalManagerTestMaxJoinExit
[PASS] test_maxJoin_6dp_mixedCaps_differentGroups() (gas: 3611117)
[PASS] test_maxJoin_6dp_mixedCaps_mixedGroup() (gas: 3091555)
[PASS] test_maxJoin_6dp_mixedCaps_sameGroup() (gas: 3525927)
[PASS] test_maxJoin_maxCaps_differentGroups() (gas: 3550608)
[PASS] test_maxJoin_maxCaps_mixedGroup() (gas: 3043412)
[PASS] test_maxJoin_maxCaps_sameGroup() (gas: 3465687)
[PASS] test_maxJoin_mixedCaps_differentGroups() (gas: 3610275)
[PASS] test_maxJoin_mixedCaps_mixedGroup() (gas: 3090434)
[PASS] test_maxJoin_mixedCaps_sameGroup() (gas: 3525021)
[PASS] test_maxJoin_zeroCaps_differentGroups() (gas: 3132417)
[PASS] test_maxJoin_zeroCaps_mixedGroup() (gas: 2692218)
```

```
[PASS] test_maxJoin_zeroCaps_sameGroup() (gas: 3047463)
Suite result: ok. 12 passed; 0 failed; 0 skipped; finished in 39.97ms (63.30ms CPU time)

Ran 10 tests for test/foundry/unit/investments/opal/OpalManager.t.sol:OpalManagerTestExit
[PASS] test_exit_equalSplit_bsDoesntAddUp_leftOver() (gas: 3701947)
[PASS] test_exit_equalSplit_bsDoesntAddUp_underflow() (gas: 3635710)
[PASS] test_exit_equalSplit_success() (gas: 3675952)
[PASS] test_exit_failBadBalanceSheetData() (gas: 16544)
[PASS] test_exit_failNoAdapters() (gas: 19938)
[PASS] test_exit_failWrongNumAllocations() (gas: 604398)
[PASS] test_exit_failWrongNumAssetsOrLiabilities() (gas: 647307)
[PASS] test_exit_skewedSplit() (gas: 3675907)
[PASS] test_exit_zeroAssetsOrLiabilities() (gas: 3476606)
[PASS] test_exit_zeroExistingBalances() (gas: 2955156)
Suite result: ok. 10 passed; 0 failed; 0 skipped; finished in 39.81ms (69.17ms CPU time)

Ran 12 tests for test/foundry/unit/investments/opal/OpalManager.t.sol:OpalManagerTestAddAdapters
[PASS] test_addAdapter_fail_badImpl() (gas: 174079)
[PASS] test_addAdapter_fail_bothAssetAndLiability() (gas: 2174147)
[PASS] test_addAdapter_fail_duplicateTokens() (gas: 1167232)
[PASS] test_addAdapter_fail_max_adapters() (gas: 8177795)
[PASS] test_addAdapter_fail_noAssetsOrLiabilities() (gas: 200749)
[PASS] test_addAdapter_success_noAssets() (gas: 459132)
[PASS] test_addAdapter_success_noLiabilities() (gas: 453663)
[PASS] test_addAdapter_success_oneAdapter() (gas: 987974)
[PASS] test_addAdapter_success_threeAdapters() (gas: 2918284)
[PASS] test_addAdapter_success_twoDistinctAdapters() (gas: 2034561)
[PASS] test_addAdapter_success_twoFullyOverlappingAdapters() (gas: 1743458)
[PASS] test_addAdapter_success_twoPartiallyOverlappingAdapters() (gas: 1796088)
Suite result: ok. 12 passed; 0 failed; 0 skipped; finished in 40.15ms (30.54ms CPU time)

Ran 13 tests for test/foundry/unit/investments/opal/OpalManager.t.sol:OpalManagerTestJoin
[PASS] test_allocateAcrossAdapters_revert() (gas: 21791)
[PASS] test_allocateAcrossAdapters_view() (gas: 5263465)
[PASS] test_allocateAcrossAdapters_zeroCombinedBal() (gas: 2917372)
[PASS] test_join_equalSplit_bsDoesntAddUp_leftOver() (gas: 3694083)
[PASS] test_join_equalSplit_bsDoesntAddUp_underflow() (gas: 3633355)
[PASS] test_join_equalSplit_success() (gas: 3667939)
[PASS] test_join_failBadBalanceSheetData() (gas: 16568)
[PASS] test_join_failNoAdapters() (gas: 19960)
[PASS] test_join_failWrongNumAllocations() (gas: 604378)
[PASS] test_join_failWrongNumAssetsOrLiabilities() (gas: 647214)
[PASS] test_join_skewedSplit() (gas: 3668068)
[PASS] test_join_zeroAssetsOrLiabilities() (gas: 3476373)
[PASS] test_join_zeroExistingBalances() (gas: 2954947)
Suite result: ok. 13 passed; 0 failed; 0 skipped; finished in 40.16ms (90.45ms CPU time)

Ran 10 tests for test/foundry/unit/investments/opal/OpalManager.t.sol:OpalManagerTestAccess
[PASS] test_addAdapter_access() (gas: 18461)
[PASS] test_exit_access() (gas: 22362)
[PASS] test_join_access() (gas: 22383)
[PASS] test_multicall_access() (gas: 16100)
[PASS] test_recoverToken_access() (gas: 17962)
[PASS] test_removeAdapter_access() (gas: 17834)
[PASS] test_setFees_access() (gas: 15818)
[PASS] test_setPaused_access() (gas: 13455)
[PASS] test_setPauser_access() (gas: 17938)
[PASS] test_setPluginApproved_access() (gas: 17959)
Suite result: ok. 10 passed; 0 failed; 0 skipped; finished in 40.13ms (5.60ms CPU time)

Ran 3 tests for test/foundry/unit/investments/opal/OpalManagerBundler.t.sol:OpalManagerTestBundler
[PASS] test_bundle_supply() (gas: 287026)
[PASS] test_bundle_withCallback_condensed() (gas: 548858)
[PASS] test_bundle_withCallback_expanded() (gas: 598512)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 11.46ms (4.38ms CPU time)

Ran 2 tests for test/foundry/unit/investments/opal/OpalVaultMaxJoinExit.t.sol:OpalVaultMaxJoinExitNoAdaptersTest
[PASS] test_maxExit() (gas: 88079)
[PASS] test_maxJoin() (gas: 92775)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 16.43ms (1.80ms CPU time)

Ran 2 tests for test/foundry/unit/investments/opal/OpalVaultMaxJoinExit.t.sol:
OpalVaultMaxJoinExitComplexWithOverlappingSeparateGroupIdTest
[PASS] test_maxExitWithToken_noSupplyCap_withAdapterCap() (gas: 913635)
[PASS] test_maxJoinWithToken_noSupplyCap_withAdapterCap() (gas: 917350)
```

```
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 40.52ms (22.32ms CPU time)

Ran 2 tests for
  ↳ test/foundry/unit/investments/opal/OpalVaultMaxJoinExit.t.sol:OpalVaultMaxJoinExitComplexNonOverlappingTest
[PASS] test_maxExitWithToken_noSupplyCap_withAdapterCap() (gas: 1335422)
[PASS] test_maxJoinWithToken_noSupplyCap_withAdapterCap() (gas: 1339725)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 46.32ms (32.93ms CPU time)

Ran 2 tests for test/foundry/unit/investments/opal/OpalVaultMaxJoinExit.t.sol:
OpalVaultMaxJoinExitComplexWithOverlappingPartialGroupIdTest
[PASS] test_maxExitWithToken_noSupplyCap_withAdapterCap() (gas: 1045668)
[PASS] test_maxJoinWithToken_noSupplyCap_withAdapterCap() (gas: 1049857)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 46.34ms (24.62ms CPU time)

Ran 6 tests for test/foundry/unit/investments/opal/OpalVaultMaxJoinExit.t.sol:OpalVaultMaxJoinExit6dpAssetTest
[PASS] test_fuzz_maxExitWithToken_noSupplyCap_withAdapterCap(uint256,uint256) (runs: 512, : 275707, ~: 275675)
[PASS] test_fuzz_maxJoinWithToken_noSupplyCap_withAdapterCap(uint256,uint256) (runs: 512, : 278816, ~: 278816)
[PASS] test_maxExitWithToken_noSupplyCap_withAdapterCap() (gas: 340676)
[PASS] test_maxExitWithToken_withSupplyCap_withAdapterCap() (gas: 342641)
[PASS] test_maxJoinWithToken_noSupplyCap_withAdapterCap() (gas: 344413)
[PASS] test_maxJoinWithToken_withSupplyCap_withAdapterCap() (gas: 351112)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 1.22s (2.40s CPU time)

Ran 6 tests for
  ↳ test/foundry/unit/investments/opal/OpalVaultMaxJoinExit.t.sol:OpalVaultMaxJoinExitBasic18dpWithShareFeesTest
[PASS] test_fuzz_maxExitWithToken_noSupplyCap_withAdapterCap(uint256,uint256) (runs: 512, : 278103, ~: 278173)
[PASS] test_fuzz_maxJoinWithToken_noSupplyCap_withAdapterCap(uint256,uint256) (runs: 512, : 280697, ~: 280769)
[PASS] test_maxExitWithToken_noSupplyCap_withAdapterCap() (gas: 344329)
[PASS] test_maxExitWithToken_withSupplyCap_withAdapterCap() (gas: 346114)
[PASS] test_maxJoinWithToken_noSupplyCap_withAdapterCap() (gas: 347091)
[PASS] test_maxJoinWithToken_withSupplyCap_withAdapterCap() (gas: 354570)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 1.23s (2.43s CPU time)

Ran 6 tests for test/foundry/unit/investments/opal/OpalVaultMaxJoinExit.t.sol:OpalVaultMaxJoinExit6dpLiabilityTest
[PASS] test_fuzz_maxExitWithToken_noSupplyCap_withAdapterCap(uint256,uint256) (runs: 512, : 275730, ~: 275739)
[PASS] test_fuzz_maxJoinWithToken_noSupplyCap_withAdapterCap(uint256,uint256) (runs: 512, : 278736, ~: 278704)
[PASS] test_maxExitWithToken_noSupplyCap_withAdapterCap() (gas: 341126)
[PASS] test_maxExitWithToken_withSupplyCap_withAdapterCap() (gas: 342617)
[PASS] test_maxJoinWithToken_noSupplyCap_withAdapterCap() (gas: 344539)
[PASS] test_maxJoinWithToken_withSupplyCap_withAdapterCap() (gas: 351088)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 1.23s (2.43s CPU time)

Ran 10 tests for test/foundry/unit/investments/opal/OpalVaultMaxJoinExit.t.sol:OpalVaultMaxJoinExitBasic18dpTest
[PASS] test_fuzz_maxExitWithToken_noSupplyCap_withAdapterCap(uint256,uint256) (runs: 512, : 275709, ~: 275761)
[PASS] test_fuzz_maxJoinWithToken_noSupplyCap_withAdapterCap(uint256,uint256) (runs: 512, : 278818, ~: 278837)
[PASS] test_maxExitWithToken_noSupplyCap_noAdapterCap() (gas: 187498)
[PASS] test_maxExitWithToken_noSupplyCap_withAdapterCap() (gas: 340700)
[PASS] test_maxExitWithToken_noSupplyCap_zeroAvailableCap() (gas: 197223)
[PASS] test_maxExitWithToken_withSupplyCap_withAdapterCap() (gas: 342617)
[PASS] test_maxJoinWithToken_noSupplyCap_noAdapterCap() (gas: 191682)
[PASS] test_maxJoinWithToken_noSupplyCap_withAdapterCap() (gas: 343963)
[PASS] test_maxJoinWithToken_noSupplyCap_zeroAvailableCap() (gas: 199711)
[PASS] test_maxJoinWithToken_withSupplyCap_withAdapterCap() (gas: 351088)
Suite result: ok. 10 passed; 0 failed; 0 skipped; finished in 1.24s (2.39s CPU time)

Ran 12 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterAaveV3.t.sol:OpalAdapterAaveV3TestAccess
[PASS] test_access_borrow() (gas: 22189)
[PASS] test_access_claimAllRewards() (gas: 29416)
[PASS] test_access_exit() (gas: 23967)
[PASS] test_access_join() (gas: 23914)
[PASS] test_access_repay() (gas: 22747)
[PASS] test_access_setEModeCategory() (gas: 18399)
[PASS] test_access_setMaxLoanUtilizationRatioOnJoin() (gas: 18383)
[PASS] test_access_setReferralCode() (gas: 18445)
[PASS] test_access_setUserUseReserveAsCollateral() (gas: 20719)
[PASS] test_access_supplyCollateral() (gas: 22894)
[PASS] test_access_updateAavePool() (gas: 18255)
[PASS] test_access_withdrawCollateral() (gas: 23376)
Suite result: ok. 12 passed; 0 failed; 0 skipped; finished in 837.13ms (3.00ms CPU time)

Ran 7 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterAaveV3.t.sol:OpalAdapterAaveV3TestViews
[PASS] test_currentLtv_multiAsset() (gas: 1049860)
[PASS] test_currentLtv_noCollateral() (gas: 43110)
[PASS] test_currentLtv_noDebt() (gas: 371988)
[PASS] test_exitMetrics_repay() (gas: 1065134)
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[PASS] test_exitMetrics_withdraw() (gas: 1046731)
[PASS] test_joinMetrics_borrow() (gas: 1263104)
[PASS] test_joinMetrics_supply() (gas: 1169440)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 906.34ms (308.05ms CPU time)

Ran 20 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterAaveV3.t.sol:OpalAdapterAaveV3TestJoinExit
[PASS] test_exit_fail_notEnough() (gas: 42309)
[PASS] test_exit_fail_wrongLength() (gas: 27830)
[PASS] test_exit_maxWithdraw() (gas: 1333412)
[PASS] test_exit_repayTooMuch() (gas: 1333432)
[PASS] test_exit_someRepay_zeroWithdraw() (gas: 1181695)
[PASS] test_exit_success() (gas: 1650798)
[PASS] test_exit_withdrawTooMuch() (gas: 1189196)
[PASS] test_exit_zeroRepay_someWithdrawal() (gas: 1206473)
[PASS] test_exit_zeroRepay_zeroWithdraw() (gas: 1245013)
[PASS] test_join_fail_noAllowance() (gas: 28235)
[PASS] test_join_fail_noBalance() (gas: 60367)
[PASS] test_join_fail_notEnoughCollateral() (gas: 956170)
[PASS] test_join_fail_overSafeLtv() (gas: 1799708)
[PASS] test_join_fail_overUtilizationRatio_withBorrow() (gas: 1033984)
[PASS] test_join_fail_overUtilizationRatio_zeroBorrow() (gas: 733423)
[PASS] test_join_fail_wrongLength() (gas: 27767)
[PASS] test_join_someCollateral_zeroBorrow() (gas: 524705)
[PASS] test_join_success() (gas: 1244216)
[PASS] test_join_zeroCollateral_someBorrow() (gas: 940716)
[PASS] test_join_zeroCollateral_zeroBorrow() (gas: 141576)
Suite result: ok. 20 passed; 0 failed; 0 skipped; finished in 908.95ms (1.04s CPU time)

Ran 10 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterAaveV3.t.sol:OpalAdapterAaveV3TestMaxJoinExit
[PASS] test_maxExit_borrowCaps_liability() (gas: 998433)
[PASS] test_maxExit_default() (gas: 981872)
[PASS] test_maxExit_supplyCaps_asset() (gas: 998417)
[PASS] test_maxExit_zeroFromFlags_asset() (gas: 1019022)
[PASS] test_maxExit_zeroFromFlags_liability() (gas: 1019084)
[PASS] test_maxJoin_borrowCaps_liability() (gas: 1060793)
[PASS] test_maxJoin_default() (gas: 1054934)
[PASS] test_maxJoin_supplyCaps_asset() (gas: 1060871)
[PASS] test_maxJoin_zeroFromFlags_asset() (gas: 1101637)
[PASS] test_maxJoin_zeroFromFlags_liability() (gas: 1101087)
Suite result: ok. 10 passed; 0 failed; 0 skipped; finished in 1.10s (1.12s CPU time)

Ran 26 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterAaveV3.t.sol:OpalAdapterAaveV3TestAdmin
[PASS] test_balanceSheet_collateralDonationsAreCounted() (gas: 360138)
[PASS] test_balanceSheet_noDirectDonations() (gas: 401470)
[PASS] test_encodeImmutableArgs_fail_badTokens() (gas: 234599)
[PASS] test_encodeImmutableArgs_fail_noAssets() (gas: 14127)
[PASS] test_encodeImmutableArgs_fail_sameToken() (gas: 52098)
[PASS] test_encodeImmutableArgs_fail_tooManyAssets() (gas: 14889)
[PASS] test_initialization() (gas: 415198)
[PASS] test_initialize_fail_badAToken() (gas: 316428)
[PASS] test_initialize_fail_badDToken() (gas: 201583)
[PASS] test_initialize_fail_maxURTooHigh() (gas: 226299)
[PASS] test_initialize_fail_missingCollateralToken() (gas: 310095)
[PASS] test_initialize_fail_noAssets() (gas: 237384)
[PASS] test_initialize_fail_sameAsset() (gas: 230902)
[PASS] test_initialize_fail_tooManyAssets() (gas: 240308)
[PASS] test_initialize_zeroEmode() (gas: 423601)
[PASS] test_positionDetailsWithMixedCollateralNoEmode() (gas: 755611)
[PASS] test_positionDetailsWithPtSusdeCollateralNoEmode() (gas: 446703)
[PASS] test_positionDetailsWithSusdeCollateralNoEmode() (gas: 466222)
[PASS] test_setEModeCategory() (gas: 1847893)
[PASS] test_setMaxLoanUtilizationRatioOnJoin_failure() (gas: 16391)
[PASS] test_setMaxLoanUtilizationRatioOnJoin_success() (gas: 24482)
[PASS] test_setReferralCode_success() (gas: 24366)
[PASS] test_setUserUseReserveAsCollateral_fail() (gas: 57551)
[PASS] test_setUserUseReserveAsCollateral_success() (gas: 354184)
[PASS] test_updateAavePool() (gas: 158313)
[PASS] test_updateAavePool_fail_sameAddress() (gas: 23733)
Suite result: ok. 26 passed; 0 failed; 0 skipped; finished in 1.10s (1.10s CPU time)

Ran 21 tests for
↳ test/foundry/unit/investments/opal/adapters/OpalAdapterAaveV3.t.sol:OpalAdapterAaveV3TestBundlerActions
[PASS] test_borrow_notEnoughCollateral() (gas: 562097)
[PASS] test_borrow_success() (gas: 686552)
[PASS] test_borrow_zeroAmount() (gas: 357816)
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[PASS] test_claimAllRewards() (gas: 945938)
[PASS] test_repay_exact() (gas: 746723)
[PASS] test_repay_max_overMinAmount() (gas: 842675)
[PASS] test_repay_max_underMinAmount() (gas: 823500)
[PASS] test_repay_tooManyAssets() (gas: 840987)
[PASS] test_supplyCollateral_maxOverMinAmount() (gas: 345878)
[PASS] test_supplyCollateral_maxUnderMinAmount() (gas: 236027)
[PASS] test_supplyCollateral_nonBSAsset() (gas: 456016)
[PASS] test_supplyCollateral_specifiedAmt() (gas: 357149)
[PASS] test_supplyCollateral_zeroAmt() (gas: 134076)
[PASS] test_validateLoanUtilizationRatioInRange() (gas: 121859)
[PASS] test_validateLoanUtilizationRatioInRange_badParam() (gas: 11418)
[PASS] test_validateLtvInRange() (gas: 787245)
[PASS] test_validateLtvInRange_badParam() (gas: 37360)
[PASS] test_withdrawCollateral_max() (gas: 403920)
[PASS] test_withdrawCollateral_notEnough() (gas: 78264)
[PASS] test_withdrawCollateral_specifiedAmt() (gas: 408656)
[PASS] test_withdrawCollateral_zeroAmt() (gas: 78216)
Suite result: ok. 21 passed; 0 failed; 0 skipped; finished in 1.60s (1.77s CPU time)

Ran 3 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterBase.t.sol:OpalAdapterBaseTestAccess
[PASS] test_access_erc20Approve() (gas: 26484)
[PASS] test_access_merk1Claim() (gas: 26851)
[PASS] test_access_setDeprecated() (gas: 18382)
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 7.44ms (724.95µs CPU time)

Ran 6 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterBase.t.sol:OpalAdapterBaseTestBundlerActions
[PASS] test_erc20Approve_failBadToken() (gas: 20759)
[PASS] test_erc20Approve_success() (gas: 244160)
[PASS] test_merk1Claim_differentRecipient() (gas: 71525)
[PASS] test_merk1Claim_no_inputs() (gas: 24868)
[PASS] test_merk1Claim_sameRecipient() (gas: 62981)
[PASS] test_merk1Claim_zeroAmount() (gas: 80076)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 7.77ms (2.89ms CPU time)

Ran 11 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterBase.t.sol:OpalAdapterBaseTestAdmin
[PASS] test_description() (gas: 205955)
[PASS] test_impl_construction() (gas: 60791)
[PASS] test_initialization() (gas: 62844)
[PASS] test_initialize_fail_alreadyInitialized() (gas: 354656)
[PASS] test_initialize_fail_badBundler() (gas: 330181)
[PASS] test_initialize_fail_badInitArgs() (gas: 118937)
[PASS] test_initialize_fail_badOwner() (gas: 121622)
[PASS] test_initialize_fail_noInit() (gas: 2729346)
[PASS] test_initialize_impl_fail() (gas: 10969)
[PASS] test_initialize_success() (gas: 358985)
[PASS] test_setDeprecated() (gas: 27039)
Suite result: ok. 11 passed; 0 failed; 0 skipped; finished in 7.48ms (4.31ms CPU time)

Ran 9 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterEuler.t.sol:OpalAdapterEulerTestAccess
[PASS] test_access_borrow() (gas: 23161)
[PASS] test_access_exit() (gas: 24116)
[PASS] test_access_join() (gas: 24108)
[PASS] test_access_repay() (gas: 22820)
[PASS] test_access_setMaxLoanUtilizationRatioOnJoin() (gas: 18329)
[PASS] test_access_supplyCollateral() (gas: 22866)
[PASS] test_access_withDeferredEulerChecks_bundler_check() (gas: 51813)
[PASS] test_access_withDeferredEulerChecks_evc_check() (gas: 222423)
[PASS] test_access_withdrawCollateral() (gas: 23174)
Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 624.91ms (3.13ms CPU time)

Ran 2 tests for
↳ test/foundry/unit/investments/opal/adapters/OpalAdapterEuler.t.sol:OpalAdapterEulerTestEulerPauseMethods
[PASS] test_euler_global_pause() (gas: 946744430)
[PASS] test_euler_hook_revert() (gas: 609053)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 636.98ms (8.13ms CPU time)

Ran 9 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterEuler.t.sol:OpalAdapterEulerTestViews
[PASS] test_areAnyVaultOperationsDisabled() (gas: 132638)
[PASS] test_currentLtv_noCollateral() (gas: 33289)
[PASS] test_currentLtv_noDebt() (gas: 444205)
[PASS] test_currentLtv_withDebt() (gas: 729258)
[PASS] test_exitMetrics_repay() (gas: 755483)
[PASS] test_exitMetrics_withdraw() (gas: 755929)
[PASS] test_joinMetrics_borrow() (gas: 860935)
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[PASS] test_joinMetrics_supply() (gas: 817715)
[PASS] test_positionDetails() (gas: 768665)
Suite result: ok. 9 passed; 0 failed; 0 skipped; finished in 653.14ms (67.06ms CPU time)

Ran 11 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterEuler.t.sol:OpalAdapterEulerTestAdmin
[PASS] test_encodeImmutableArgs_fail_sameAsset() (gas: 26303)
[PASS] test_initialize_fail_zero_address_loanToken() (gas: 160828)
[PASS] test_initialization_euler_adapter() (gas: 432375)
[PASS] test_initialize_fail_badCollateralToken() (gas: 225405)
[PASS] test_initialize_fail_badLoanToken() (gas: 160784)
[PASS] test_initialize_fail_collateral_loan_tokens_check() (gas: 104480)
[PASS] test_initialize_fail_maxURTooHigh() (gas: 144100)
[PASS] test_initialize_fail_missingBorrowVault() (gas: 148219)
[PASS] test_initialize_fail_zero_address_collateral() (gas: 225342)
[PASS] test_setMaxLoanUtilizationRatioOnJoin_failure() (gas: 16329)
[PASS] test_setMaxLoanUtilizationRatioOnJoin_success() (gas: 24387)
Suite result: ok. 11 passed; 0 failed; 0 skipped; finished in 655.38ms (11.07ms CPU time)

Ran 5 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterEuler.t.sol:OpalAdapterEulerTestMaxJoinExit
[PASS] test_maxExit_default() (gas: 717633)
[PASS] test_maxJoin_borrowCaps_liability() (gas: 782082)
[PASS] test_maxJoin_default() (gas: 764633)
[PASS] test_maxJoin_supplyCaps_asset() (gas: 784996)
[PASS] test_maxJoin_zeroFromFlags_asset() (gas: 849469)
Suite result: ok. 5 passed; 0 failed; 0 skipped; finished in 655.38ms (53.67ms CPU time)

Ran 20 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterEuler.t.sol:OpalAdapterEulerTestJoinExit
[PASS] test_exit_fail_notEnough() (gas: 38731)
[PASS] test_exit_fail_wrongLength() (gas: 37776)
[PASS] test_exit_maxWithdraw() (gas: 1040248)
[PASS] test_exit_repayTooMuch() (gas: 1059014)
[PASS] test_exit_someRepay_zeroWithdraw() (gas: 826459)
[PASS] test_exit_success() (gas: 1384722)
[PASS] test_exit_withdrawTooMuch() (gas: 929082)
[PASS] test_exit_zeroRepay_someWithdrawal() (gas: 1035050)
[PASS] test_exit_zeroRepay_zeroWithdraw() (gas: 1055558)
[PASS] test_join_fail_noAllowance() (gas: 29593)
[PASS] test_join_fail_noBalance() (gas: 56980)
[PASS] test_join_fail_notEnoughCollateral() (gas: 1233030)
[PASS] test_join_fail_overSafeLtv() (gas: 688044)
[PASS] test_join_fail_overUtilizationRatio_withBorrow() (gas: 753404)
[PASS] test_join_fail_overUtilizationRatio_zeroBorrow() (gas: 392056)
[PASS] test_join_fail_wrongLength() (gas: 37769)
[PASS] test_join_someCollateral_zeroBorrow() (gas: 642471)
[PASS] test_join_success() (gas: 1071753)
[PASS] test_join_zeroCollateral_someBorrow() (gas: 647674)
[PASS] test_join_zeroCollateral_zeroBorrow() (gas: 190922)
Suite result: ok. 20 passed; 0 failed; 0 skipped; finished in 657.22ms (138.05ms CPU time)

Ran 20 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterEuler.t.sol:OpalAdapterEulerTestBundlerActions
[PASS] test_borrow_notEnoughCollateral() (gas: 606419)
[PASS] test_borrow_success() (gas: 773802)
[PASS] test_borrow_zeroAmount() (gas: 293527)
[PASS] test_repay_exact() (gas: 791976)
[PASS] test_repay_max_overMinAmount() (gas: 916048)
[PASS] test_repay_max_underMinAmount() (gas: 895374)
[PASS] test_repay_tooManyAssets() (gas: 914119)
[PASS] test_supplyCollateral_maxOverMinAmount() (gas: 290780)
[PASS] test_supplyCollateral_maxUnderMinAmount() (gas: 232446)
[PASS] test_supplyCollateral_specifiedAmt() (gas: 317835)
[PASS] test_supplyCollateral_zeroAmt() (gas: 84378)
[PASS] test_validateLoanUtilizationRatioInRange_badParam() (gas: 11386)
[PASS] test_validateLoanUtilizationRatioInRange_rangeChecks() (gas: 119088)
[PASS] test_validateLtvInRange() (gas: 1025095)
[PASS] test_validateLtvInRange_badParam() (gas: 30019)
[PASS] test_withDeferredEulerChecks_withdrawThenRepay() (gas: 1089091)
[PASS] test_withdrawCollateral_max() (gas: 398800)
[PASS] test_withdrawCollateral_notEnough() (gas: 88446)
[PASS] test_withdrawCollateral_specifiedAmt() (gas: 398605)
[PASS] test_withdrawCollateral_zeroAmt() (gas: 16842)
Suite result: ok. 20 passed; 0 failed; 0 skipped; finished in 658.61ms (102.38ms CPU time)

Ran 4 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterMorpho.t.sol:OpalAdapterMorphoTestMaxJoinExit
[PASS] test_maxExit_default() (gas: 378251)
[PASS] test_maxJoin_default() (gas: 378652)
```

```
[PASS] test_maxJoin_lowMaxUr() (gas: 386527)
[PASS] test_maxJoin_noMaxUr() (gas: 385255)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 544.97ms (11.59ms CPU time)

Ran 7 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterMorpho.t.sol:OpalAdapterMorphoTestViews
[PASS] test_currentLoanTokenUtilizationRatio_noSupply() (gas: 459942)
[PASS] test_currentLtv_noDebt() (gas: 249644)
[PASS] test_currentLtv_withDebt() (gas: 296687)
[PASS] test_currentLtv_zeroOraclePrice() (gas: 293585)
[PASS] test_joinMetrics_noMaxUr() (gas: 36159)
[PASS] test_liquidationLtv() (gas: 8454)
[PASS] test_positionDetails() (gas: 514060)
Suite result: ok. 7 passed; 0 failed; 0 skipped; finished in 550.09ms (10.95ms CPU time)

Ran 10 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterMorpho.t.sol:OpalAdapterMorphoTestAdmin
[PASS] test_balanceSheet_collateralDonationsAreCounted() (gas: 253869)
[PASS] test_balanceSheet_noDirectDonations() (gas: 361909)
[PASS] test_encodeImmutableArgs_fail_sameAsset() (gas: 11902)
[PASS] test_initialization() (gas: 169658)
[PASS] test_initialize_fail() (gas: 465144)
[PASS] test_initialize_fail_sameTokens() (gas: 115351)
[PASS] test_setMaxLoanUtilizationRatioOnJoin_failure() (gas: 16373)
[PASS] test_setMaxLoanUtilizationRatioOnJoin_success() (gas: 24463)
[PASS] test_setMaxSafeLtv_failure() (gas: 16293)
[PASS] test_setMaxSafeLtv_success() (gas: 24601)
Suite result: ok. 10 passed; 0 failed; 0 skipped; finished in 550.17ms (7.49ms CPU time)

Ran 10 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterMorpho.t.sol:OpalAdapterMorphoTestAccess
[PASS] test_access_borrow() (gas: 23225)
[PASS] test_access_exit() (gas: 23922)
[PASS] test_access_join() (gas: 23980)
[PASS] test_access_onMorphoRepay() (gas: 21632)
[PASS] test_access_onMorphoSupplyCollateral() (gas: 21557)
[PASS] test_access_repay() (gas: 23430)
[PASS] test_access_setMaxLoanUtilizationRatioOnJoin() (gas: 18427)
[PASS] test_access_setMaxSafeLtv() (gas: 18339)
[PASS] test_access_supplyCollateral() (gas: 23199)
[PASS] test_access_withdrawCollateral() (gas: 23123)
Suite result: ok. 10 passed; 0 failed; 0 skipped; finished in 551.08ms (2.28ms CPU time)

Ran 20 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterMorpho.t.sol:OpalAdapterMorphoTestJoinExit
[PASS] test_exit_fail_notEnough() (gas: 26248)
[PASS] test_exit_fail_overSafeLtv() (gas: 618808)
[PASS] test_exit_fail_wrongLength() (gas: 24950)
[PASS] test_exit_repayTooMuch() (gas: 679148)
[PASS] test_exit_someRepay_zeroWithdraw() (gas: 531359)
[PASS] test_exit_success() (gas: 710936)
[PASS] test_exit_withdrawTooMuch() (gas: 578640)
[PASS] test_exit_zeroRepay_someWithdrawal() (gas: 465329)
[PASS] test_exit_zeroRepay_zeroWithdraw() (gas: 414435)
[PASS] test_join_fail_noAllowance() (gas: 26302)
[PASS] test_join_fail_noBalance() (gas: 60584)
[PASS] test_join_fail_overLiquidationLtv() (gas: 294060)
[PASS] test_join_fail_overSafeLtv() (gas: 351136)
[PASS] test_join_fail_overUtilizationRatio_withBorrow() (gas: 364613)
[PASS] test_join_fail_overUtilizationRatio_zeroBorrow() (gas: 307876)
[PASS] test_join_fail_wrongLength() (gas: 25020)
[PASS] test_join_someCollateral_zeroBorrow() (gas: 351447)
[PASS] test_join_success() (gas: 436233)
[PASS] test_join_zeroCollateral_someBorrow() (gas: 352520)
[PASS] test_join_zeroCollateral_zeroBorrow() (gas: 270913)
Suite result: ok. 20 passed; 0 failed; 0 skipped; finished in 551.15ms (47.94ms CPU time)

Ran 36 tests for
↳ test/foundry/unit/investments/opal/adapters/OpalAdapterMorpho.t.sol:OpalAdapterMorphoTestBundlerActions
[PASS] test_borrow_assets() (gas: 332252)
[PASS] test_borrow_assets_and_shares() (gas: 25039)
[PASS] test_borrow_notEnoughCollateral() (gas: 250526)
[PASS] test_borrow_shares() (gas: 332183)
[PASS] test_borrow_zeroAmount() (gas: 228425)
[PASS] test_onMorphoRepay_withCallback_failBundlerUnset() (gas: 15831)
[PASS] test_onMorphoSupplyCollateral_badCalldata() (gas: 15048)
[PASS] test_onMorphoSupplyCollateral_withCallback_failBundlerUnset() (gas: 15807)
[PASS] test_repay_assets_and_shares() (gas: 285431)
[PASS] test_repay_assets_exactAssets() (gas: 331615)
```

```
[PASS] test_repay_assets_maxAssets_overThreshold() (gas: 313733)
[PASS] test_repay_assets_maxAssets_underThreshold() (gas: 332520)
[PASS] test_repay_assets_partialAssets() (gas: 334100)
[PASS] test_repay_assets_tooManyAssets() (gas: 489693)
[PASS] test_repay_assets_withCallback_2Deep() (gas: 752576)
[PASS] test_repay_assets_withCallback_failWhenCalledDirectly() (gas: 298294)
[PASS] test_repay_shares_exactShares() (gas: 481807)
[PASS] test_repay_shares_maxShares_noDebt() (gas: 269211)
[PASS] test_repay_shares_maxShares_withDebt() (gas: 484608)
[PASS] test_repay_shares_partialShares() (gas: 328775)
[PASS] test_repay_shares_tooManyShares() (gas: 286864)
[PASS] test_supplyCollateral_max_overThreshold() (gas: 237653)
[PASS] test_supplyCollateral_max_underThreshold() (gas: 203309)
[PASS] test_supplyCollateral_specifiedAmt() (gas: 236818)
[PASS] test_supplyCollateral_withCallback_1Deep() (gas: 317460)
[PASS] test_supplyCollateral_withCallback_2Deep() (gas: 498265)
[PASS] test_supplyCollateral_withCallback_failWhenCalledDirectly() (gas: 214848)
[PASS] test_supplyCollateral_zeroAmt() (gas: 54174)
[PASS] test_validateLoanUtilizationRatioInRange_badParam() (gas: 11408)
[PASS] test_validateLoanUtilizationRatioInRange_rangeChecks() (gas: 42884)
[PASS] test_validateLtvInRange() (gas: 386126)
[PASS] test_validateLtvInRange_badParam() (gas: 13569)
[PASS] test_withdrawCollateral_max() (gas: 273770)
[PASS] test_withdrawCollateral_notEnough() (gas: 28567)
[PASS] test_withdrawCollateral_specifiedAmt() (gas: 271073)
[PASS] test_withdrawCollateral_zeroAmt() (gas: 25199)
Suite result: ok. 36 passed; 0 failed; 0 skipped; finished in 551.22ms (59.80ms CPU time)
```

```
Ran 2 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterSpotAssets.t.sol:OpalAdapterSpotAssetsTestAccess
[PASS] test_access_exit() (gas: 23886)
[PASS] test_access_join() (gas: 23900)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 7.47ms (610.13µs CPU time)
```

```
Ran 8 tests for test/foundry/unit/investments/opal/adapters/OpalAdapterSpotAssets.t.sol:OpalAdapterSpotAssetsTestAdmin
[PASS] test_adapterInit_failTooManyAssets() (gas: 165233)
[PASS] test_adapterInit_failZeroAssetAddr() (gas: 115878)
[PASS] test_adapterInit_failZeroAssets() (gas: 111025)
[PASS] test_balanceSheet() (gas: 224450)
[PASS] test_encodeImmutableArgs_failTooManyAssets() (gas: 12550)
[PASS] test_encodeImmutableArgs_failZeroAssetAddr() (gas: 11782)
[PASS] test_encodeImmutableArgs_failZeroAssets() (gas: 11286)
[PASS] test_initialization() (gas: 75721)
Suite result: ok. 8 passed; 0 failed; 0 skipped; finished in 8.93ms (2.68ms CPU time)
```

```
Ran 11 tests for
↳ test/foundry/unit/investments/opal/adapters/OpalAdapterSpotAssets.t.sol:OpalAdapterSpotAssetsTestJoinExit
[PASS] test_exit_allZeroAmount() (gas: 293101)
[PASS] test_exit_fail_notEnough() (gas: 26248)
[PASS] test_exit_fail_wrongLength() (gas: 28776)
[PASS] test_exit_success() (gas: 389473)
[PASS] test_exit_zeroAmount() (gas: 318027)
[PASS] test_join_allZeroAmount() (gas: 54042)
[PASS] test_join_fail_noAllowance() (gas: 26452)
[PASS] test_join_fail_noBalance() (gas: 56161)
[PASS] test_join_fail_wrongLength() (gas: 28641)
[PASS] test_join_success() (gas: 289901)
[PASS] test_join_zeroAmount() (gas: 203610)
Suite result: ok. 11 passed; 0 failed; 0 skipped; finished in 8.95ms (6.39ms CPU time)
```

8.2 Automated Tools

8.2.1 AuditAgent

The AuditAgent is an AI-powered smart contract auditing tool that analyses code, detects vulnerabilities, and provides actionable fixes. It accelerates the security analysis process, complementing human expertise with advanced AI models to deliver efficient and comprehensive smart contract audits. Available at <https://app.auditagent.nethermind.io>.

9 About Nethermind

Nethermind is a Blockchain Research and Software Engineering company. Our work touches every part of the web3 ecosystem - from layer 1 and layer 2 engineering, cryptography research, and security to application-layer protocol development. We offer strategic support to our institutional and enterprise partners across the blockchain, digital assets, and DeFi sectors, guiding them through all stages of the research and development process, from initial concepts to successful implementation.

We offer security audits of projects built on EVM-compatible chains and Starknet. We are active builders of the Starknet ecosystem, delivering a node implementation, a block explorer, a Solidity-to-Cairo transpiler, and formal verification tooling. Nethermind also provides strategic support to our institutional and enterprise partners in blockchain, digital assets, and decentralized finance (DeFi). In the next paragraphs, we introduce the company in more detail.

Blockchain Security: At Nethermind, we believe security is vital to the health and longevity of the entire Web3 ecosystem. We provide security services related to Smart Contract Audits, Formal Verification, and Real-Time Monitoring. Our Security Team comprises blockchain security experts in each field, often collaborating to produce comprehensive and robust security solutions. The team has a strong academic background, can apply state-of-the-art techniques, and is experienced in analyzing cutting-edge Solidity and Cairo smart contracts, such as ArgentX and StarkGate (the bridge connecting Ethereum and StarkNet). Most team members hold a Ph.D. degree and actively participate in the research community, accounting for 240+ articles published and 1,450+ citations in Google Scholar. The security team adopts customer-oriented and interactive processes where clients are involved in all stages of the work.

Blockchain Core Development: Our core engineering team, consisting of over 20 developers, maintains, improves, and upgrades our flagship product - the Nethermind Ethereum Execution Client. The client has been successfully operating for several years, supporting both the Ethereum Mainnet and its testnets, and now accounts for nearly a quarter of all synced Mainnet nodes. Our unwavering commitment to Ethereum's growth and stability extends to sidechains and layer 2 solutions. Notably, we were the sole execution layer client to facilitate Gnosis Chain's Merge, transitioning from Aura to Proof of Stake (PoS), and we are actively developing a full-node client to bolster Starknet's decentralization efforts. Our core team equips partners with tools for seamless node set-up, using generated docker-compose scripts tailored to their chosen execution client and preferred configurations for various network types.

DevOps and Infrastructure Management: Our infrastructure team ensures our partners' systems operate securely, reliably, and efficiently. We provide infrastructure design, deployment, monitoring, maintenance, and troubleshooting support, allowing you to focus on your core business operations. Boasting extensive expertise in Blockchain as a Service, private blockchain implementations, and node management, our infrastructure and DevOps engineers are proficient with major cloud solution providers and can host applications in-house or on clients' premises. Our global in-house SRE teams offer 24/7 monitoring and alerts for both infrastructure and application levels. We manage over 5,000 public and private validators and maintain nodes on major public blockchains such as Polygon, Gnosis, Solana, Cosmos, Near, Avalanche, Polkadot, Aptos, and StarkWare L2. Sedge is an open-source tool developed by our infrastructure experts, designed to simplify the complex process of setting up a proof-of-stake (PoS) network or chain validator. Sedge generates docker-compose scripts for the entire validator set-up based on the chosen client, making the process easier and quicker while following best practices to avoid downtime and being slashed.

Cryptography Research: At Nethermind, our cryptography Research team conducts cutting-edge internal research and collaborates closely with external partners on cryptographic protocols, consensus design, succinct arguments and folding schemes, elliptic curve-based STARK protocols, post-quantum security and zero-knowledge proofs (ZKPs). Our research has led to influential contributions, including Zinc (Crypto '25), Mova, FLI (Asiacrypt '24), and foundational results in Fiat-Shamir security and STARK proof batching. Complementing this theoretical work, our engineering expertise is demonstrated through implementations such as the Latticefold aggregation scheme, the Labrador proof system, zkvm-benchmarks, and Plonk Verifier in Cairo. This combined strength in theory and engineering enables us to deliver cutting-edge cryptographic solutions to partners and clients.

Smart Contract Development & DeFi Research: Our smart contract development and DeFi research team comprises 40+ world-class engineers who collaborate closely with partners to identify needs and work on value-adding projects. The team specializes in Solidity and Cairo development, architecture design, and DeFi solutions, including DEXs, AMMs, structured products, derivatives, and money market protocols, as well as ERC20, 721, and 1155 token design. Our research and data analytics focuses on three key areas: technical due diligence, market research, and DeFi research. Utilizing a data-driven approach, we offer in-depth insights and outlooks on various industry themes.

Our suite of L2 tooling: Warp is Starknet's approach to EVM compatibility. It allows developers to take their Solidity smart contracts and transpile them to Cairo, Starknet's smart contract language. In the short time since its inception, the project has accomplished many achievements, including successfully transpiling Uniswap v3 onto Starknet using Warp.

- **Voyager** is a user-friendly Starknet block explorer that offers comprehensive insights into the Starknet network. With its intuitive interface and powerful features, Voyager allows users to easily search for and examine transactions, addresses, and contract details. As an essential tool for navigating the Starknet ecosystem, Voyager is the go-to solution for users seeking in-depth information and analysis;
- **Horus** is an open-source formal verification tool for StarkNet smart contracts. It simplifies the process of formally verifying Starknet smart contracts, allowing developers to express various assertions about the behavior of their code using a simple assertion language;
- **Juno** is a full-node client implementation for Starknet, drawing on the expertise gained from developing the Nethermind Client. Written in Golang and open-sourced from the outset, Juno verifies the validity of the data received from Starknet by comparing it to proofs retrieved from Ethereum, thus maintaining the integrity and security of the entire ecosystem.

General Advisory to Clients

As auditors, we recommend that any changes or updates made to the audited codebase undergo a re-audit or security review to address potential vulnerabilities or risks introduced by the modifications. By conducting a re-audit or security review of the modified codebase, you can significantly enhance the overall security of your system and reduce the likelihood of exploitation. However, we do not possess the authority or right to impose obligations or restrictions on our clients regarding codebase updates, modifications, or subsequent audits. Accordingly, the decision to seek a re-audit or security review lies solely with you.

Disclaimer

This report is based on the scope of materials and documentation provided by you to [Nethermind](#) in order that [Nethermind](#) could conduct the security review outlined in **1. Executive Summary** and **2. Audited Files**. The results set out in this report may not be complete nor inclusive of all vulnerabilities. [Nethermind](#) has provided the review and this report on an as-is, where-is, and as-available basis. You agree that your access and/or use, including but not limited to any associated services, products, protocols, platforms, content, and materials, will be at your sole risk. Blockchain technology remains under development and is subject to unknown risks and flaws. The review does not extend to the compiler layer, or any other areas beyond the programming language, or other programming aspects that could present security risks. This report does not indicate the endorsement of any particular project or team, nor guarantee its security. No third party should rely on this report in any way, including for the purpose of making any decisions to buy or sell a product, service or any other asset. To the fullest extent permitted by law, [Nethermind](#) disclaims any liability in connection with this report, its content, and any related services and products and your use thereof, including, without limitation, the implied warranties of merchantability, fitness for a particular purpose, and non-infringement. [Nethermind](#) does not warrant, endorse, guarantee, or assume responsibility for any product or service advertised or offered by a third party through the product, any open source or third-party software, code, libraries, materials, or information linked to, called by, referenced by or accessible through the report, its content, and the related services and products, any hyperlinked websites, any websites or mobile applications appearing on any advertising, and [Nethermind](#) will not be a party to or in any way be responsible for monitoring any transaction between you and any third-party providers of products or services. As with the purchase or use of a product or service through any medium or in any environment, you should use your best judgment and exercise caution where appropriate. FOR AVOIDANCE OF DOUBT, THE REPORT, ITS CONTENT, ACCESS, AND/OR USAGE THEREOF, INCLUDING ANY ASSOCIATED SERVICES OR MATERIALS, SHALL NOT BE CONSIDERED OR RELIED UPON AS ANY FORM OF FINANCIAL, INVESTMENT, TAX, LEGAL, REGULATORY, OR OTHER ADVICE.